

A unique positive completion with a missing central block

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Conjecture 00000008590

Claim addressed. The source asserts that the minimal positive completion is unique if and only if the missing block lies at a corner. We disprove the necessity direction for a fixed 3×3 block layout, using scalar blocks on the real Hilbert space $\mathbb{R}^3$. The example covers both minimum operator norm and minimality in Loewner order.

Proposition 1. *For the partial operator matrix*

$$P = \begin{pmatrix} 0 & 0 & 0 \\ 0 & ? & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

the zero operator is the unique completion of minimum operator norm and the unique Loewner-minimal positive completion. The unspecified block is central and is not a corner.

Proof. We use the standard Euclidean inner product, its orthonormal coordinate basis, and the induced operator norm. A positive operator here means a self-adjoint operator T satisfying $\langle x, Tx \rangle \geq 0$ for every x . Its matrix entries are $\langle e_i, Te_j \rangle$, and all entries except $(i, j) = (1, 1)$, with indices $0, 1, 2$, must equal zero.

The zero operator is a positive completion. Every operator has nonnegative norm, so the minimum completion norm is 0. If a completion attains this norm, the norm's definiteness gives $T = 0$. Thus the norm-minimizing completion is unique.

For completeness, Loewner order is $S \preceq T$ if $\langle x, Sx \rangle \leq \langle x, Tx \rangle$ for every x . Every positive completion satisfies $0 \preceq T$, so zero is a least positive completion. If T is Loewner-minimal among these completions, the admissibility of zero implies $T \preceq 0$. Positivity then yields

$$\langle x, Tx \rangle = 0 \quad \text{for every } x.$$

Self-adjointness and polarization imply $T = 0$. Explicitly, evaluating the preceding identity at $x + Tx$ and expanding gives

$$0 = \langle x, Tx \rangle + \langle x, T(Tx) \rangle + \langle Tx, Tx \rangle + \langle Tx, T(Tx) \rangle = 2\langle Tx, Tx \rangle.$$

Hence $Tx = 0$ for every x . Conversely zero is Loewner-minimal because it is least. This proves uniqueness under this interpretation as well.

In the fixed layout, a corner has both row and column indices in $\{0, 2\}$. The missing index $(1, 1)$ satisfies neither endpoint condition. This contradicts the asserted necessity of a corner. $\square$

Scope and formalization. All specified blocks are allowed to be zero; neither language imposes a nondegeneracy hypothesis. The block position is understood in the given layout, as required by a condition about corners. The other positive completions are $\text{diag}(0, t, 0)$ for $t \geq 0$, although their classification is unnecessary for the proof. Lean uses actual continuous linear operators on `EuclideanSpace` $\mathbb{R}$ (Fin 3), actual inner-product matrix entries and operator norms. It proves zero's feasibility and leastness, and proves that every norm-minimal or Loewner-minimal feasible operator equals zero. The polarization calculation is also formalized. We do not address the source's other completion claims.